25900565 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEPHYA00 - INTRODUCTION TO MEDICAL PHYSICS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **2** marks

1. Describe the concept of the color triangle in relation to cone excitation. [U] / [CO1]
2. State the differences between continuous wave (CW) Doppler and pulsed wave (PW) Doppler. [K] / [CO3]
3. Name any two precordial leads used in ECG. [K] / [CO6]
4. List two uses of photodynamic therapy in eye treatment. [K] / [CO1]
5. State the formula for acoustic impedance in terms of density and speed of sound. [K] / [CO3]
6. Explain why scotopic (night) vision images appear bluish-green. [U] / [CO1]
7. Describe two clinical uses of speech audiometry. [K] / [CO4]
8. Explain the function of the eustachian tube in hearing. [U] / [CO4]
9. List the different regions of the electromagnetic spectrum in order of increasing frequency. [K] / [CO1]
10. Explain the parts of a transducer. [U] / [CO5]

11. Define an endoscope and mention its basic function. [K] / [CO2]

12. Explain briefly EEG. [K] / [CO6]

13. Define frequency of sound wave and state its SI unit. [K] / [CO3]

14. What is the principle of CT imaging? [K] / [CO6]

[2x10 = 20]

Part B

Short Essay Type Questions

Answer any **6** questions

Each question carries **5** marks

15. Explain the function of the interferometer in OCT. [K] / [CO2]

16. Explain the mechanism of light dispersion with the help of a prism [U] / [CO1]

17. Describe the input characteristics of a transistor in CB mode. [K] / [CO5]

18. Differentiate between CO_2, Nd:YAG, and excimer lasers in terms of wavelength and applications. [U] / [CO1]

19. · Describe briefly the process of attenuation of absorption. [K] / [CO3]

20. What are the properties of X-rays that make them useful in medical imaging? [K] / [CO5]

21. Why are op-amps widely used in biomedical instruments? [U] / [CO5]

22. Describe the functioning of electromagnetic tracking systems in freehand ultrasound. [K] / [CO4]

[5x6 = 30]